Implicit Neural Representations with Periodic Activation Functions

Machine Learning in Physics

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Background / Introduction

Implicit Neural Representations (INRs) use neural networks to represent continuous physical fields by mapping coordinates directly to quantities like intensity or potential. The SIREN model introduced sinusoidal activation functions, enabling smooth and accurate modeling of both signals and their derivatives (Sitzmann et al. [2020]). This makes SIRENs especially suitable for solving and analyzing physical systems governed by partial differential equations (PDEs).

The network architecture creates an implicit representation of signals (audio, video, image) by creating a function $\Phi_\theta(x)$ which minimizes a loss function of the original signal. This neural network is a fully connected perceptron that uses sines for its activation functions: $\phi_i(x) = sin(W_i x + b_i)$. In the paper, the authors used 5-6 layer MLPs with 256-1024 hidden features and a linear output layer. Critical for training was the initialization (which sped up convergence) of the layers, where the first layer was initialized uniformly randomly and then scaled by a frequency ω_0. Other layers of size n are initialized uniformly randomly as well with the form $W_i \sim U(-\sqrt{6/n}, \sqrt{6/n})$, which ensures stable gradient propagation for periodic activation. Finally, the training proceeds by sampling coordinates via Monte Carlo sampling at every iteration (pixel coordinates for images and interior/boundary points for PDEs). This ensures unbiased approximation of the continuous loss-function defined on the domain. Using autodiff, one can also implement other re-constructions where the gradient $\nabla\Phi_\theta$ or laplacian $\nabla^2\Phi_\theta$ can be supervised to minimize loss functions derived from PDEs.

Here we reproduce the main results of the SIREN paper and extend them to other images and PDEs. We find that the method outlined in the paper works but can be both slow and sensitive to the hyperparameters selected for training. In Van Gogh's *Starry Night*, we find that periodic activation functions allow the model to recapitulate turbulent signatures buried in the power spectrum of the real painting. Finally, we use a SIREN to solve the Poisson-Boltzmann Equation, finding that the method laid out in the paper generalizes well to another PDE.

Reproducing the results of the paper

Representing images with SIREN

The most basic use of SIREN is in creating neural representations for images. Here we reproduce the SIREN representation of the classic cameraman image using two methods. First, the model is fit directly using the MSE loss against the ground truth (Fig. 1A). Second, the model is fit using the MSE loss of the *gradient* of the image, again evaluated against the (gradient of the) ground truth image (Fig. 1B). Each task requires less than 500 epochs of training (around 1 minute) to produce visually accurate results.

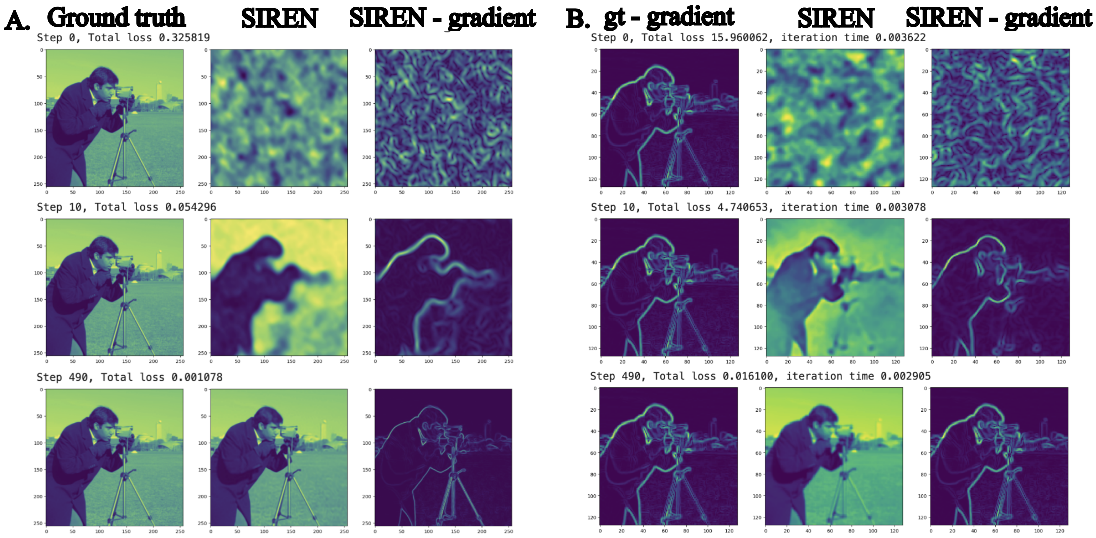


Figure 1: Creating an implicit neural representation of an image with SIREN. A. Training a SIREN using the MSE loss function on the ground truth $y(\mathbf{x})$: $\mathcal{L} = \int_\Omega (\Phi(\mathbf{x}) - y(\mathbf{x}))^2$. B. Training a SIREN using the MSE loss on the gradient of the ground truth $\nabla y(\mathbf{x})$: $\mathcal{L} = \int_\Omega |\nabla\Phi(\mathbf{x}) - \nabla y(\mathbf{x})|^2$.

Solving the Helmholtz Equation

In Sitzmann et.al. Sitzmann et al. [2020], one of the interesting applications of SIREN is in solving physically-motivated PDEs such as the 2D Helmholtz Equation. The aim is to train a neural network using the residual of the PDE as the loss function:

$$\mathcal{L} = \int_\Omega \lambda(\mathbf{x}) \|\nabla^2\Phi(\mathbf{x}) + \frac{1}{c(\mathbf{x})^2}\omega^2\Phi(\mathbf{x}) + s(\mathbf{x})\| \mathrm{d}\mathbf{x} \tag{1}$$

where the model, $\Phi(\mathbf{x})$, is a complex-valued function of position $\mathbf{x}$, $s(\mathbf{x})$ is a source term, and $c(\mathbf{x})$ is the wave speed, set to 1 throughout the domain. $\lambda(\mathbf{x})$ is a hyperparameter which weighs points where $s(\mathbf{x}) \neq 0$ relative to those with $s(\mathbf{x}) = 0$. For simplicity, the authors use a gaussian source positioned at the center of the domain. To prevent boundary effects, the authors employ a perfectly matched layer (PML) which damps the solution near the boundary. This requires a modified form of the Helmholtz Equation presented above:

$$\begin{aligned} \frac{\partial}{\partial x_1}\left(\frac{e_{x_2}}{e_{x_1}}\frac{\partial\Phi(\mathbf{x})}{\partial x_1}\right) + \frac{\partial}{\partial x_2}\left(\frac{e_{x_1}}{e_{x_2}}\frac{\partial\Phi(\mathbf{x})}{\partial x_2}\right) + e_{x_1}e_{x_2}\left(\frac{\omega}{c}\right)^2\Phi(\mathbf{x}) + s(\mathbf{x}) = 0 \\ e_{x_i} = \begin{cases} 1 - ia_0\left(\frac{\ell_{x_i}}{L_{\mathrm{PML}}}\right)^2, & \text{if } x_i \in \partial\Omega \\ 1, & \text{otherwise} \end{cases} \end{aligned} \tag{2}$$

This version of the equation accounts for spatial variation in the wave speed, which is induced via an imaginary term in the permittivity e_{x_i}. The strength of damping depends on another hyperparameter a_0, and the distance ℓ_{x_i} from and size of L_{PML} the dissipative layer. The authors take $L_{\mathrm{PML}} = 0.5$ on a domain which ranges from -1 to 1 in both x and y. The actual neural network architecture is a simple fully-connected multi-layer perceptron with 4 hidden layers each containing 256 nodes.

Ultimately we were unable to totally replicate the results from the paper, though our results generally agree up to some further tweaking of training parameters, inputs to the loss function, etc. For example, the authors did not specify the particular image size, magnitude of the source $s(\mathbf{x})$, or frequency ω that they used. They also did not provide any quantitative comparisons between their trained models and the ground truth, nor did they present any loss curves to show the progression of the training. The size and strength of the dissipative layer turned out to be important, as the authors' cited value of $a_0 = 5$ was insufficient to suppress boundary effects

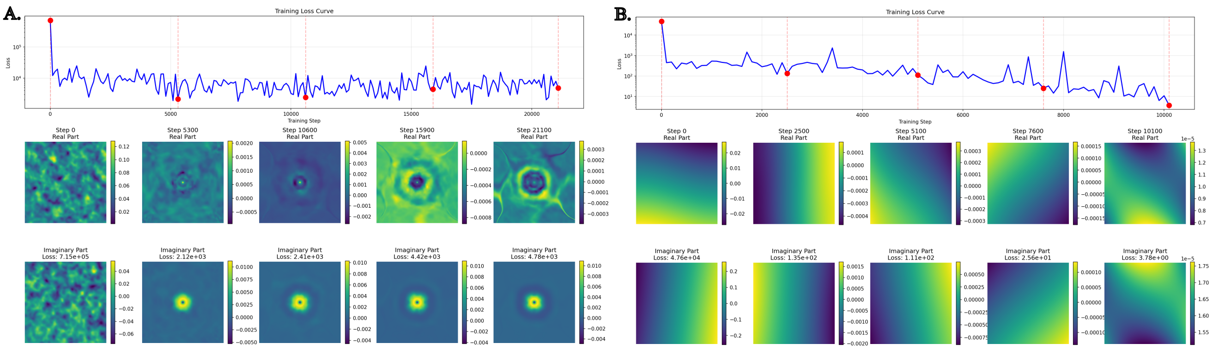


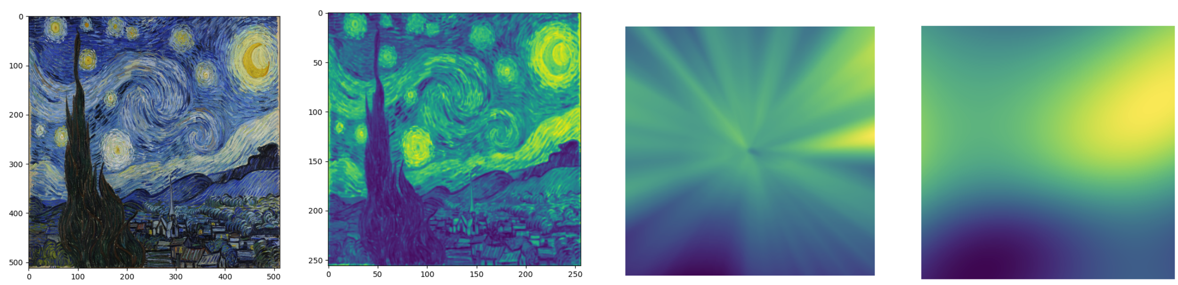
Figure 2: Training progression for the Helmholtz Equation (Eq. 2) using a SIREN (sine activation function, A) and a normal MLP (tanh activation function, B).

in our training. We also were unable to run the training for as long as the authors did (50,000 epochs) due to hardware and time constraints. The training progression of our best models using sinusoidal and tanh activation functions are presented in Fig. 2A-B. A ReLU-based model (not shown) did not perform any better than the tanh-based model. Interestingly, the ReLU and tanh models each reach a lower value of the loss than the SIREN, but appear visually less accurate. Overall, it is visually clear that the SIREN provides a much better approximation to the correct solution (Fig. 6) than the model using tanh activations.

Extending SIREN to new applications

Representing Starry Night with SIREN

After reproducing several aspects of the paper, we extend the application to another image (for reasons that would become clear in the next section), namely that of Van Gogh's *Starry Nights*. We obtain a 256 $\times$ image from Wikipedia, and then import that as input into SIREN module. The parameters of the SIREN involve 2 dimensional input features with one dimensional output, 3 hidden layers with 256 neurons. Each hidden layer is a SineLayer i.e. with sinusoidal output with the output layer being linear. We train for a total of 500 steps, with neurons initialized according the same scheme mentioned in the paper. With Adam optimizer and a L^2 loss, we obtain a high fidelity reconstruction with lowest MSE of 3.23×10^{-2} as shown in Fig. 8a. In contrast, for ReLU (with same parameters) or TanH (with more number of hidden layers), the reconstruction is low fidelity and doesn't resemble the ground truth (i.e. the original image) at all (see Fig. 3).



Ground Truth SIREN ReLU TanH

Figure 3: Starry Night reconstruction with SIREN, ReLU and TanH architectures.

Turbulent scaling laws in Starry Nights

Van Gogh's Starry Night presents a night sky whose swirling clouds span a wide range of spatial scales and visually resemble turbulent eddies. This resemblance has been explored quantitatively:

Ma et al. [2024] analyzed the luminance power spectrum and reported a $P(k) \sim k^{-5/3}$ scaling at intermediate wavenumbers, transitioning to k^{-1} at high k. The surprising facet of this result lies in its connection to Kolmogorov's 1941, which results in energy spectrum of turbulent fluids (i.e. fluids at high Reynolds no.) being a power law with $E(k) = Ck^{-5/3}$. Ma et al. [2024] further claimed that this dual cascade scaling, namely $k^{-5/3}$ and k^{-1}, matches inertial range cascade of a passively advected scalar field (i.e. a dye particle advected by turbulent fluid) with Batchelor k^{-1} power law representative of high-Schmidt number scaling at high wavenumbers. While the validity of interpreting presence of such scaling laws as signatures of turbulence isn't completely warranted (see Bourgault et al. [2025] for a critique of this specific analysis), we investigate this scaling in SIREN architecture to check if the relative intensities and their scaling can be reproduced by the sinusoidal activation functions.

To reproduce the result of Ma et al. [2024], some extensive pre-processing is required. Some of the obstacles are:

- The presence of houses, trees and other artifacts that need to manually masked in order to analyze the swirly eddies seen in the background.
- A high resolution image (30k×24k used by Ma et al. [2024]) which requires large memory/compute overhead (720 million pixels) for memory, storage, and processing . This along with copyright/licensing restrictions make thorough analysis difficult.
- Specific methodology used for binning, windowing, normalization etc. would lead to different results for the spectra.

Despite these limitations, since SIREN reconstructs the image to high fidelity, we nevertheless simply proceed to compute the 2d power spectrum (i.e. $\|\hat{f}(k)\|^2$) of 256×256 version of the image and plot the result as a function of k. We convert the luminance of the image $L(x, y)$ in greyscale, so it has a scalar intensity which is then fourier transformed to compute $E(k) = \|L_k\|^2$. For the original image, we observe a cascade which has approximately $k^{-\frac{5}{3}}$ spectrum at small wavenumbers and deviations at large wavenumbers. We see deviations from $k^{-5/3}$ scaling at high wavenumbers in Fig. 9 when local slope dlog E/dlog k is plotted. The SIREN reconstructed image captures these deviations almost exactly (see a,b subfigures in 9), indicating that the network architecture doesn't introduce additional deviations and that deviations are artifacts of non-masking of villages, trees etc. in the original image.

To test whether finer resolution can improve this, we reduce the grid spacing and use SIREN to reconstruct a smooth interpolated image on a finer grid (1024× 1024). After plotting the spectrum in 4 and local slopes in 9, we see that the large k deviations are still present, and cannot be removed without masking/significant preprocessing.

Despite the conclusive 5/3 slope at large k, SIREN nevertheless reproduces the relative intensities of the original image. In this way, the network faithfully reproduces multiscale intensities present in the original image, while the scaling laws of Ma et al. [2024] would require substantially more pre-processing and computational power.

Solving the Poisson-Boltzmann Equation

As an extension beyond the PDEs examined in the original SIREN paper, we applied the SIREN architecture to the nonlinear Poisson–Boltzmann equation (PBE), a model widely used in electrostatics, colloidal science, and soft-matter physics. In its simplest dimensionless form, the equation reads

$$-\nabla^2 \phi(\mathbf{x}) + \kappa^2 \sinh(\phi(\mathbf{x})) = \rho(\mathbf{x}), \tag{3}$$

where ϕ is the electrostatic potential, κ sets the screening length, and $\rho(\mathbf{x})$ denotes fixed charges. To create a controlled test case, we generated synthetic "ground truth" data by numerically solving the equation on a 256×256 grid using a standard finite-difference solver.

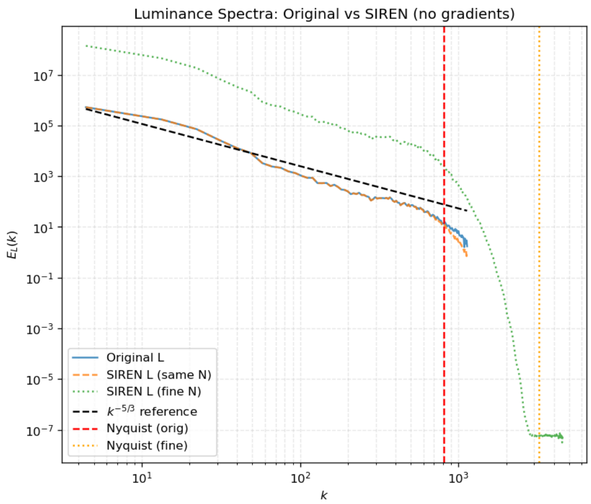


Figure 4: Plot of luminance spectra vs wavenumber (k). The plot shows spectra for the original image and SIREN reconstructed image, both on similar resolution scale. We also increase the resolution scale using SIREN and generate another spectra with finer grid spacing, shown by the green curved labeled *SIREN L (fine N)*. The nyquist wavenumber is marked, both for original and finer grid spacing.

The loss function was chosen analogously to the Helmholtz experiment, but adapted for the PB equation:

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}\left\|\nabla^2\Phi_\theta(\mathbf{x}_i) - \kappa^2\sinh(\Phi_\theta(\mathbf{x}_i))\right\|^2 + \frac{1}{M}\sum_{j=1}^{M}\|\Phi_\theta(\mathbf{x}_j) - \Phi_{\mathrm{BC}}(\mathbf{x}_j)\|^2. \quad (4)$$

The first term enforces the PDE in the interior, and the second term enforces the boundary conditions. As in the original SIREN framework, interior and boundary points were resampled using Monte Carlo sampling at every epoch. All second derivatives were computed automatically using PyTorch's higher-order autodiff. The SIREN used sine activations with the recommended initialization ($\omega_0 = 30$ on the first layer and scaled uniform weights in subsequent layers). The ReLU and tanh baselines used standard Xavier initializations (with small gains on the first layer to improve stability). Training used Adam with lr $= 10^{-4}$, cosine learning-rate annealing over 1000 steps, and gradient clipping (global norm $= 1.0$).

The three models show different behavior both numerically and visually. Key aggregated metrics are:

Model	Final training loss	MAE	Max error
SIREN	2.37×10^{-7}	7.96×10^{-5}	2.46×10^{-4}
ReLU	2.93×10^{-4}	1.81×10^{-2}	3.92×10^{-1}
Tanh	1.94×10^{-1}	1.44×10^{-2}	9.71×10^{-2}

Taken together, these numbers show that SIREN both minimizes the PDE residual to machine-precision levels and preserves the ordering and spatial structure of the ground-truth field, while ReLU provides a rough approximation and tanh fails in this setup.

The SIREN prediction virtually overlaps the finite-difference ground truth: cross-sections through the center of the domain show agreement within $\mathcal{O}(10^{-4})$. The SIREN error map is nearly zero everywhere except for slight boundary artifacts. By contrast, the ReLU model produces a much deeper and narrower potential well (overestimating the central magnitude by roughly a factor of 4), which creates a bright central error region; the tanh model collapses to an almost-constant output and therefore fails to capture the well entirely.

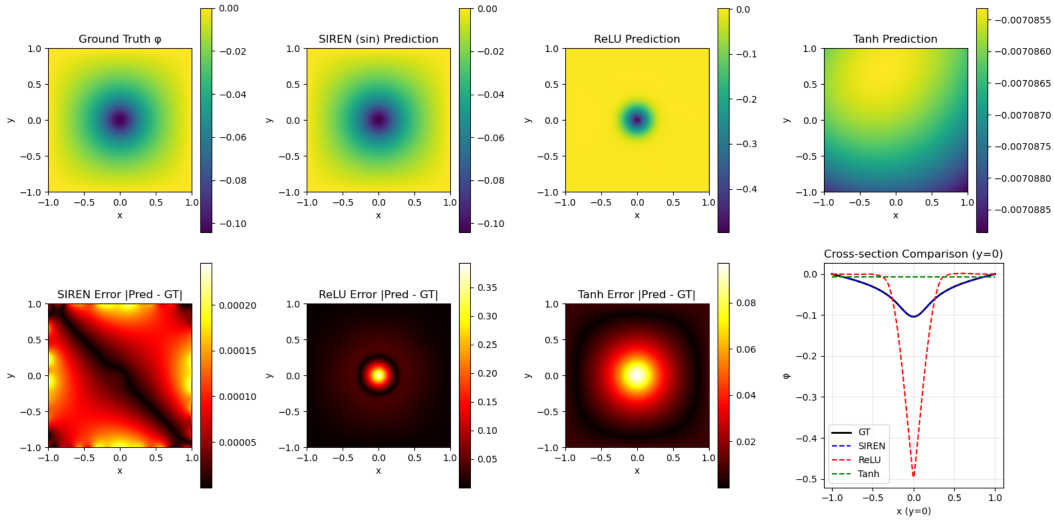


Figure 5: Predicted solutions compared to ground truth for the Poisson–Boltzmann equation.

Conclusion

In this work, we reproduced and extended the core results of SIREN-based implicit neural representations. Our experiments confirm that sinusoidal activations enable neural networks to capture fine spatial detail and accurately represent signals and their derivatives, outperforming standard ReLU and tanh architectures in both image reconstruction and PDE solving. While reproducing the Helmholtz example required additional tuning beyond what was reported in the original paper, SIRENs consistently produced the most physically meaningful solutions. We further showed that SIRENs generalize well to new domains: they preserve multiscale structure in natural images such as Starry Night and reliably solve nonlinear PDEs such as the Poisson–Boltzmann equation. Taken together, our results demonstrate that periodic activation functions provide a powerful and flexible framework for continuous representations in physics, with strong potential for future applications in scientific computing and data-driven modeling.

Acknowledgements

The link to the Github repository containing the codes used in generating plots can be found at Pascal Weisenberg [2025].

References

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Daniel Bourgault, Cédric Chavanne, Jérôme Lemelin, Sandy Grégorio, Peter S Galbraith, and Warren H Finlay. If van gogh's the starry night depicts perfect turbulence, so should degas' a woman seated beside a vase of flowers. *Bulletin of the American Meteorological Society*, 106 (8):E1703–E1707, 2025.

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Appendix

Ground Truth for the Helmholtz Equation

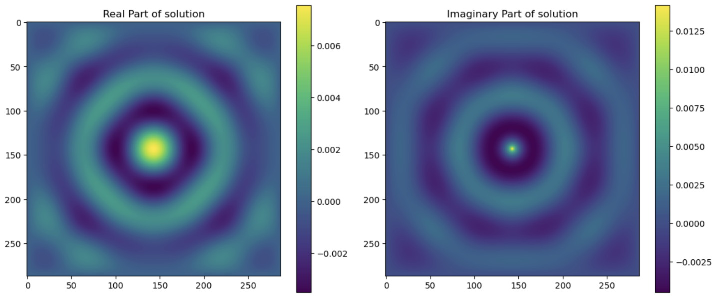


Figure 6: Solution to the Helmholtz Equation using a finite volume-based solver.

Starry Night reconstruction using SIREN

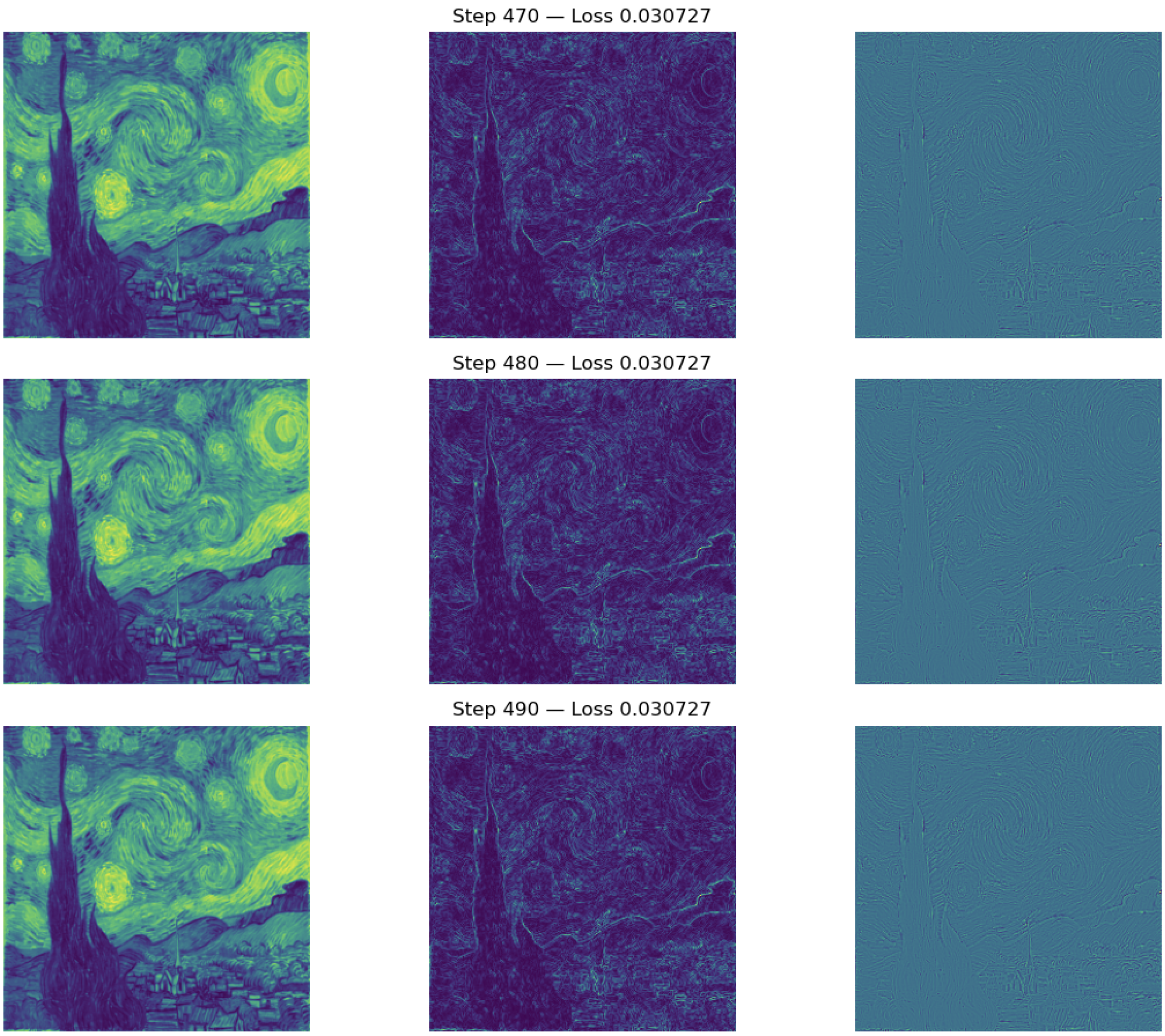
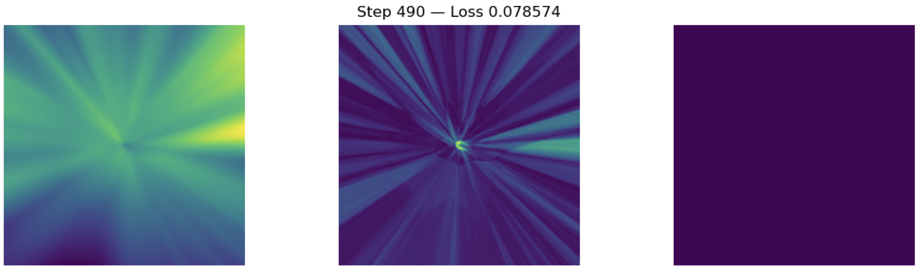
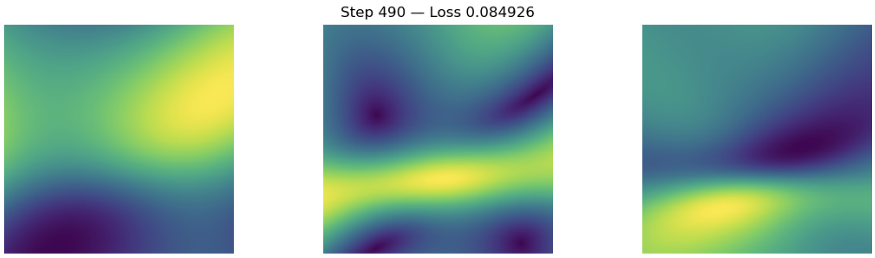


Figure 7: Starry Nights image reconstruction using SIREN. Left subfigures show image output at each epoch. Middle and right subfigures shown gradient and laplacian of the smooth SIREN interpolant respectively for the corresponding images.



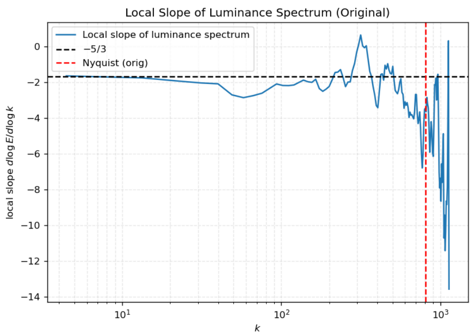
(a) Starry Nights image reconstruction using ReLU



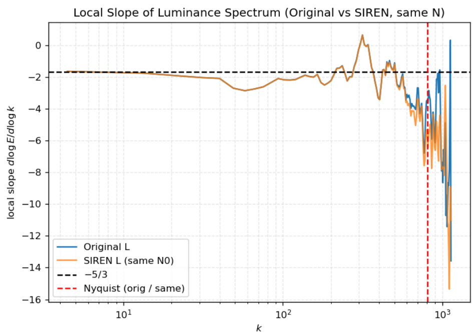
(b) Starry Nights image reconstruction using TanH

Figure 8: Starry Nights reconstruction using ReLU and TanH. Both produce poor fidelity and fail to reproduce the image.

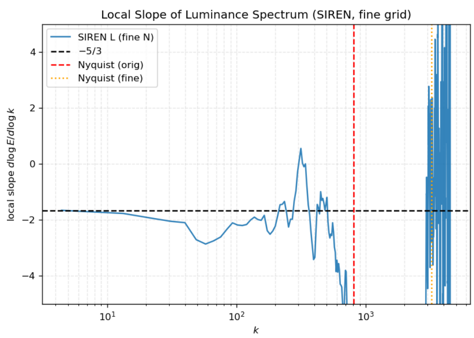
Local slopes for Starry Night using SIREN



(a) Local slope for original image



(b) Local slope for SIREN reconstructed image



(c) Local slope for SIREN reconstruction on a finer grid

Figure 9: Local slope showing scaling exponent for original, SIREN reconstructed and SIREN reconstructed with finer grid images.